Over the course of many years, Einstein mounted a series of ever more sophisticated challenges aimed at revealing gaps in the structure of quantum mechanics. One such challenge, raised in 1927 at the Fifth Physical Conference of the Solvay Institute,[8] concerns the fact that even though an electron's probability wave might look like that in Figure 4.5, whenever we measure the electron's whereabouts we always find it at one definite position or another. So, Einstein asked, doesn't that mean that the probability wave is merely a temporary stand-in for a more precise description—one yet to be discovered—that would predict the electron's position with certainty? After all, if the electron is found at X, doesn't that mean, in reality, it was at or very near X a moment before the measurement was carried out? And if so, Einstein prodded, doesn't quantum mechanics' reliance on the probability wave—a wave that, in this example, says the electron had some probability to have been far from X— reflect the theory's inadequacy to describe the true underlying reality?

Einstein's viewpoint is simple and compelling. What could be more natural than to expect a particle to be located at, or, at the very least, near where it's found a moment later? If that's the case, a deeper understanding of physics should provide *that* information and dispense with the coarser framework of probabilities. But the Danish physicist Niels Bohr and his entourage of quantum mechanics defenders disagreed. Such reasoning, they argued, is rooted in conventional thinking, according to which each electron follows a single, definite path as it wanders to and fro. And this thinking is strongly challenged by Figure 4.4, since if each electron did follow one definite path—like the classical image of a bullet fired from a gun—it would be extremely hard to explain the observed interference pattern: what would be interfering with what? Ordinary bullets fired one by one from a single gun certainly can't interfere with each other, so if electrons did travel like bullets, how would we explain the pattern in Figure 4.4?

Instead, according to Bohr and the Copenhagen interpretation of quantum mechanics he forcefully championed, *before one measures the electron's position there is no sense in even asking where it is.* It does not have a definite position. The probability wave encodes the likelihood that the electron, when examined suitably, will be found here or there, and that *truly* is all that can be said about its position. Period. The